200 max tokens, temperature=1.0, and top p=0.5. These choices were determined experimentally to encourage varied responses while maintaining coherence. Future research may seek to apply the methods presented in this paper to other prominent LLMs [21, 22] and compare results.

To compile our LLM response dataset, we assembled a grid of demographic characteristics, exhaustively capturing all possible combinations of $\langle age, nationality, sex \rangle$, $\langle age, nationality \rangle$, $\langle age, sex \rangle$, $\langle nationality, sex \rangle$, and each of $\langle age \rangle$, $\langle nationality \rangle$, $\langle sex \rangle$, given the preselected values specified above. This produced 188 demographic categories at varying levels of detail. Using the six questions listed in Table 1, we generated 1,128 unique LLM prompts. We ran each prompt 50 times to obtain multiple, varying responses, yielding a total of 56,400 LLM responses for subsequent analysis of moral value resonance.

### 2.2 Recognizing Value Resonance: Finding Implicit Values in Text

We use a transformer-based Recognizing Value Resonance (RVR) model, produced by fine-tuning a Recognizing Textual Entailment (RTE) RoBERTa model [23] on a RVR dataset [16]. The RVR model operates analogous to RTE: each LLM-generated response is a *premise*, and each of the RVR Hypotheses and Inverted RVR Hypotheses in Table 1 is a *hypothesis*. When the RVR model is given a $\langle premise, hypothesis \rangle$ pair, it produces three possible outputs: $resonance = 1$, $neutral = 0$, or $conflict = -1$.

We next projected each unique premise on to the Inglehart-Welzel traditional-secular axis [18] using a weighted sum of each premise's encoded RVR scores under each hypothesis, using the factor loadings in Table 1 from the Inglehart-Welzel factor analysis [18, 19]. This projection was calculated three times: using only the RVR hypotheses; using only the inverted hypothesis; and summing the results of these two projections when calculated using halved factor loadings to account for duplicate effect.

To compare the observed moral alignment distributions generated by the LLM to real world moral value profiles we conducted a mirrored study over the WVS. We collected real world data for our 8 nationalities of interest from the, publicly available, WVS Wave 7 cross-national dataset[24]. We focused exclusively on instances with complete demographic data on an individuals, nationality, age, sex, and complete coded responses to variables Q164, Y003, Q184, Q254, and Q45. These variables reported: "Importance of God" , "Autonomy Index" , "Justifiable: Abortion" , "National pride" , "Future changes: Greater respect for authority" . We inverted Q164 (Importance of God) which was coded with higher values indicating greater importance. The rest of the values were naturally coded with lower responses corresponding to more traditional values. We then centered each variable with its median at 0, setting the two middle integers to 0 if there were an even number of options and scaled all responses to between -1 and 1, measuring distance from "traditional" pole. Finally, as with the LLM responses, we projected each instance of the data on to the traditional-secular axis by taking a weighted sum of the coded responses using the weights from the Inglehart-Welzel factor analysis [18] reported in Table 1.

## 3 Experiments: Bias Alignment with Real-world Data

We conducted two sets of analyses to assess the alignment of biases present in LLM-generated text with real-world data for specific demographic groups. To ensure the internal validity of our experiments, we processed the LLM-generated data in three essential ways. First, we focused on LLM-generated premises that aligned with all three demographic categories of interest: Nation, Age, and Sex. This subset exclusively comprised premises generated from prompts that included information from all three categories (e.g., "You are a 35-year-old American woman"). Second, we specifically selected LLM projection scores calculated using both traditional-leaning and secular-leaning hypotheses, projected onto the axis with halved factor loadings. Our rationale for this selection is detailed in Section 5. Lastly, we exclusively included scores over premises generated when the LLM was passed the general value prompt (Table 1). This approach allowed us to better target the internal biases present in the LLMs by selecting out only the values that the LLM considered most salient to each demographic group.

In our first analysis we evaluated the extent to which LLMs successfully echoed true distributions in moral values present in demographic subsets of the population. To do this we used box and whisker plots to assess the distributional similarity between LLM and WVS-based traditional-secular

projections within each demographic category. In our second analysis, we aimed to determine the extent to which LLMs could effectively explain the underlying variance in the WVS responses by demographic delineation, using a regression analysis. We calculated mean WVS and LLM scores along the traditional-secular axis for each demographic group and regressed the two using nation-level fixed-effects.

# 4 Results

## 4.1 Value Resonance in Generated Text

Our RVR results over all LLM generated premises generally show high levels of resonance with traditional oriented hypotheses (Figure 1, “(T)"), and conflict with secular oriented hypotheses (Figure 1, “(S)"). The exceptions are hypotheses related to desired characteristics in children and abortion issues (Figure 1, 7-10).

We observe that the results do not completely mirror the hypotheses, meaning they do not entirely invert as potentially expected. This lack of mirroring is particularly pronounced in hypotheses characterized by higher levels of neutrality (lower on the y-axis). This discrepancy primarily arises from premises that exhibit non-neutrality towards either the traditional-directed or secular-directed hypotheses while remaining neutral towards the other. This likely happens due to the strong polarity of these hypotheses. Additionally, there are some instances where the same premise receives conflicting labels under opposing hypotheses. For example, the statement “God is moderately important to me." contradicts with both "God is very important in my life" and "God is irrelevant to my life." We further elaborate on these effects in Section 5.

## 4.2 Distribution Analysis against Real-World Values

Figure 2 illustrates the outcomes of our distribution analysis, which compares the moral values distributions calculated over LLM predictions to those obtained through the WVS. We have organized these comparisons by demographic characterizations: nation, age, and sex. The most striking findings emerge from the nation-based comparisons, where both WVS and LLM projections exhibit significantly higher between-groups distributional variance compared to other demographic categories. More specifically, WVS derived results demonstrate greater between-groups distributional variance than LLM predictions. Moreover, the disparity between observed LLM predictions and actual WVS scores appears to widen as WVS secularity levels increase, with the noteworthy exception of the United States. These observations suggest that LLMs encounter challenges when modeling more secular cultures, with the least alignment occurring in the most secular WVS countries, such as the Czech Republic and Japan.

By contrast, distributional differences based on age and sex, as per the WVS, exhibit considerable overlap. The LLM predictions are largely consistent with this pattern but display some discernible trends in relation to age. LLM-generated prediction distributions gradually shift towards tradition as age increases, while WVS IQRs indicate a contrary trend, indicating slight increases in secularity with age. This counterintuitive observation can be explained by a breakdown of the axis projection, revealing increased median secularity with age regarding national pride, respect for authority, and,

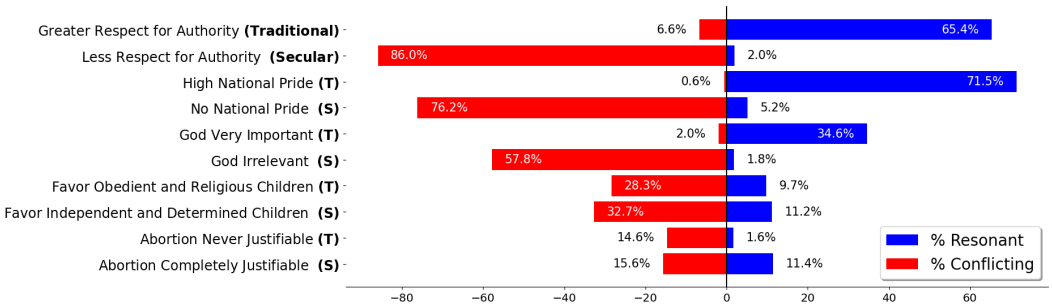


Figure 1: Waterfall plot illustrating proportion of LLM generated premises in which each hypothesis (y-axis) was non-neutral, either resonating (x-axis +) or conflicting (x-axis -)

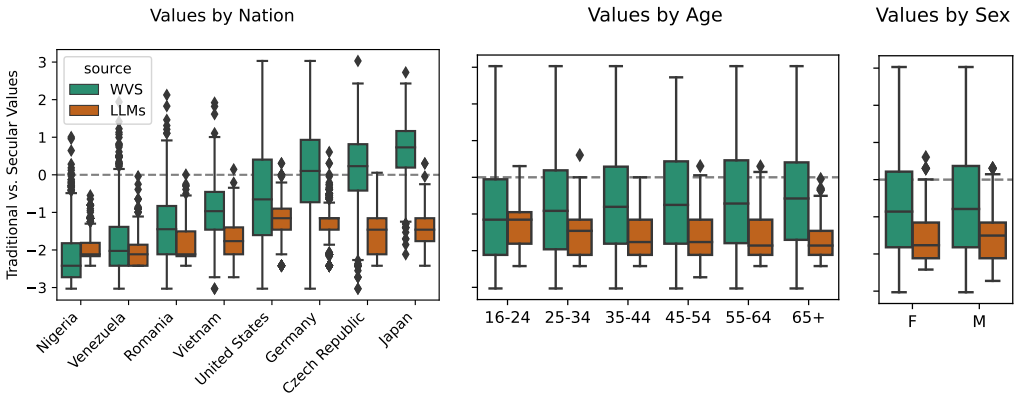


Figure 2: Boxplots comparing moral biases observed in LLM generated data with actual trends in WVS data across three demographic divides, nationality, sex, and age.

Table 2: Model fit metrics for nation-level fixed effects regression.

| | Venezuela | Nigeria | Romania | United States | Vietnam | Germany | Czech Republic | Japan |
|---|---|---|---|---|---|---|---|---|
| RMSE | 0.161 | 0.308 | 0.535 | 0.615 | 0.79 | 1.45 | 1.777 | 2.182 |
| $R^2$ | $0.44^{**}$ | 0.01 | $0.52^{***}$ | $0.8^{***}$ | 0.04 | $0.41^{**}$ | $0.57^{***}$ | 0.08 |

$^{*}$p<0.05; $^{**}$p<0.01 $^{***}$p<0.001

curiously, the importance of God. There is no clear and prominent pattern discernible between sex in either the WVS or the LLMs, although the LLMs appear to indicate that men have slightly more dispersed distributions than women, with a somewhat more secular median and a moderately lower secular skew. This higher median secularity means the LLM and WVS medians are closer for men than for women.

Generally, LLM distributions are noticeably more concentrated compared to WVS distributions, and the LLM IQRs are consistently below the WVS median scores[1] These results suggest a prevalent bias towards tradition in LLMs. However, it is important to note that, except for Germany, the Czech Republic, and Japan, the WVS distributions for all demographic groups consistently feature medians well below 0 (the traditional-secular divide).

### 4.3 Regression Analyses against Real-World Values

Figure 3 presents the outcomes of our regression analysis experiment. Focusing on the y-axis, these results echo the distribution analysis findings, highlighting that LLMs struggle to distinguish nations effectively (e.g., Japan and Germany). Moreover, the y-axis consistently reflects negative (traditional) values and greater concentration compared to the x-axis, indicating a challenge in capturing the full range of actual values and a pronounced traditional bias.

However, when considering the results introduced by national fixed effects, our regression analyses reveal intriguing insights. First and unsurprisingly, the LLM demonstrates the highest explanatory power in the United States, successfully explaining approximately 80% of the variance in WVS projections. More notably, the LLM achieves the second-highest explanatory power in the Czech Republic despite also exhibiting the second-highest error rate (Table 2). While only ever explaining between 41% and 52% of the variance in actual WVS projected values, the fixed effects regression consistently demonstrates highly significant explainable variance for Romania ($p < 0.001$) and moderately significant explainable variance for Venezuela and Germany ($p < 0.01$). The LLM projections, however, show no significant explanatory power in Japan, Vietnam, or, most notably, Nigeria, where it showed the second-lowest error rate (Table 2).

[1]Excepting Nigeria, Venezuela, and persons age 16-24.